

Topic: Database Tuning

Q. What is database Tuning?

⇒ Database tuning is a group of activities used to optimize the performance of a database.

Goal:

- o To minimize use of system resources
- o To perform task efficiently
- o To work rapidly as possible

Q. Why and when should one tune?

⇒ Slow physical I/O

⇒ Excessive CPU usage

⇒ Latch contention

Q. Causes for low performance?

⇒ Bad connection management

⇒ Bad SQL

⇒ Redo log setup problems

⇒ High amounts of recursive SQL

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Q. Where should we do tuning?

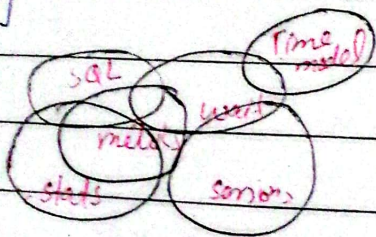
- ⇒ Database design
- ⇒ Application Tuning
- ⇒ memory Tuning
 - by properly size database buffers

U. By locking wait events, buffer hit ratios, system swapping and paging etc.

- ⇒ Tune the OS
 - cpu, I/O & memory utilization

Q. What is database statistics?

Database statistics provide information on the type of load on the db, as well as the internal and external resources used by the db.



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Q. What is AWR?

- ⇒ The automatic workload repository collects, processes, and maintains performance statistics for problem detection and self-tuning purposes.

- ⇒ This data is stored both in memory and in the database.

- ⇒ AWR automatically generates snapshots of the performance data and collects the statistics in the workload repository.